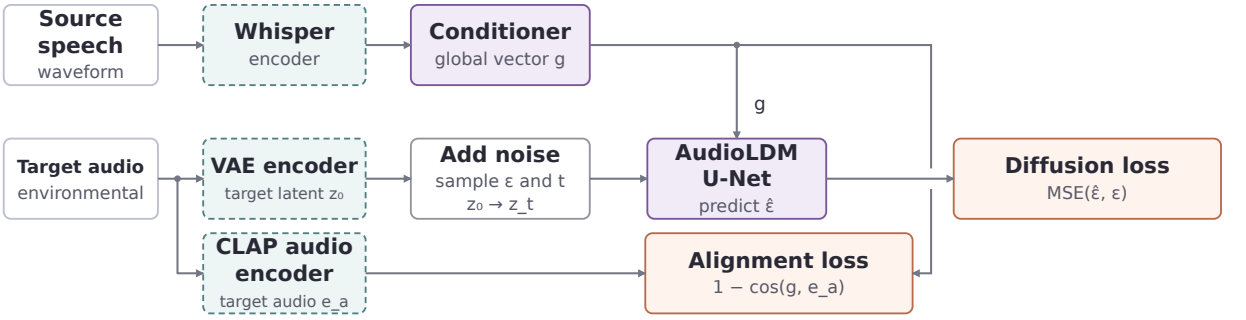
## Training the diffusion renderers

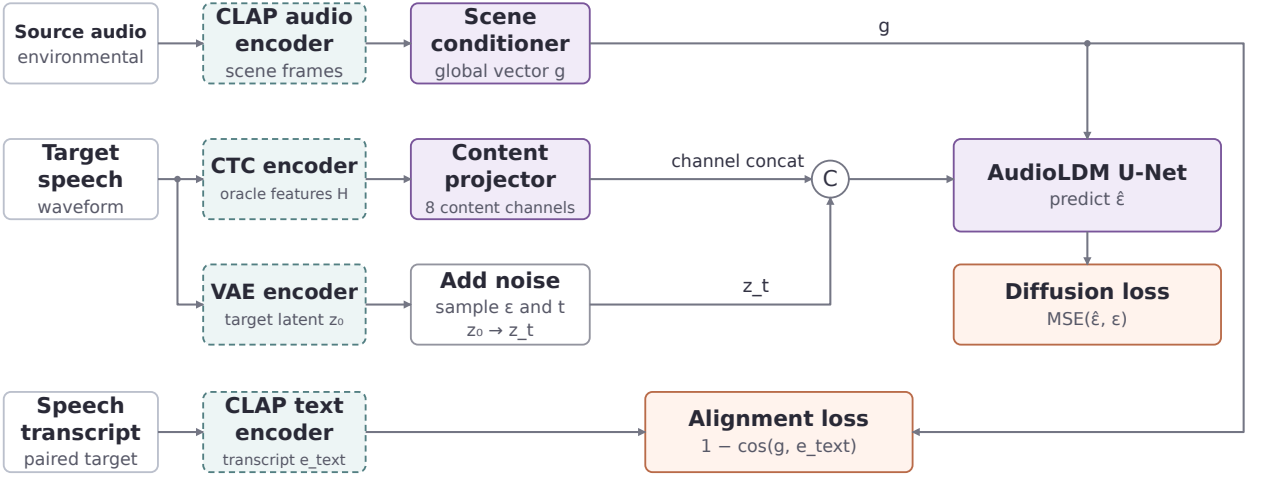
Learned / fine-tuned · Frozen · Fixed operation

### (a) Speech → audio: paired audio supplies the diffusion target




Optimize conditioner + U-Net with diffusion MSE + cosine alignment. Whisper, CLAP and VAE stay frozen.

### (b) Audio → speech: train the renderer with oracle speech content




Optimize scene conditioner + projector + U-Net. The CTC encoder, VAE and CLAP towers stay frozen.